Virtual Server Monitoring & Performance Analysis

DATA ENGINEER USE CASE ASSIGNMENT

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Tools Used: Python | Pandas | Power BI

Problem Statement

- Organizations rely on **virtual servers** to run applications and services.
- Monitoring **server performance** helps detect system issues and maintain infrastructure stability..
- This project analyzes key server metrics:
  CPU utilization, **memory usage**, disk I/O, and **network traffic**
- The objective is to understand **server health** and resource utilization patterns

Data Processing

- Data preprocessing performed using **Python and Pandas**
- Dataset explored using **df.info()** and **df.describe()**
- **Log Timestamp** converted to datetime format
- Created new column **Server_Status** based on CPU utilization
- Cleaned dataset exported for **Power BI visualization**

Healthy → CPU ≤ 70%

Warning → 70% – 85%

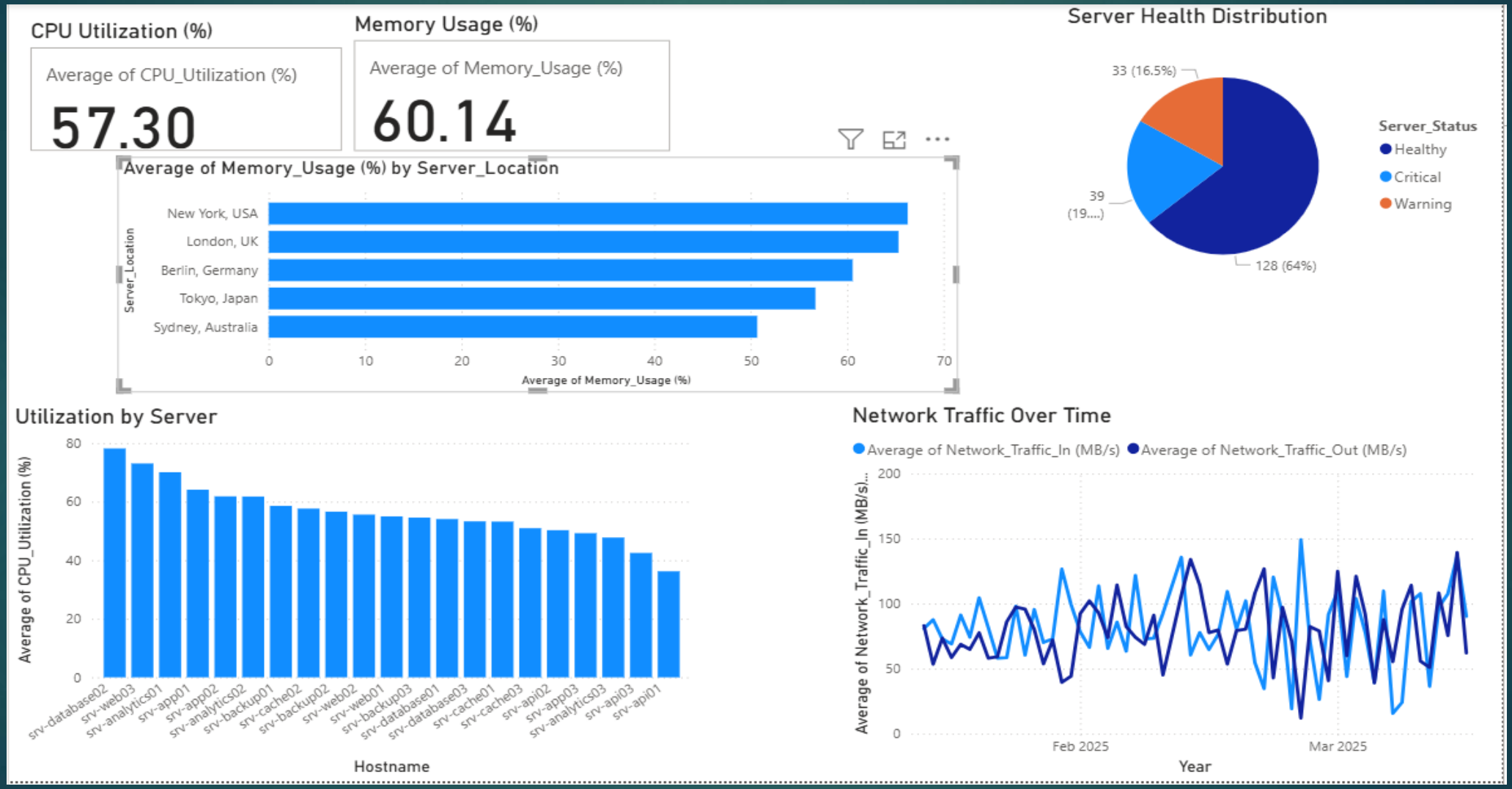
Critical → CPU > 85%

Key Performance Metrics

The system monitors the following key server performance metrics:

- **CPU Utilization (%)** – Indicates how much processor capacity is being used.
- **Memory Usage (%)** – Measures RAM consumption by running processes.
- **Disk I/O (%)** – Tracks disk read and write operations.
- **Network Traffic (MB/s)** – Monitors incoming and outgoing data flow

Power BI Dashboard



Key Insights from Dashboard

Key insights observed from the Power BI dashboard:

- Average CPU utilization across servers is around **57%**, indicating moderate system load.
- Average memory usage is approximately **60%**, showing balanced RAM utilization.
- Most servers are categorized as **Healthy**, while a smaller portion falls into Warning and Critical states.
- Memory usage varies by **server location**, with some regions showing higher utilization.
- Network traffic trends show fluctuations over time, indicating varying workload patterns

Conclusion

Server monitoring is essential for maintaining infrastructure reliability.

Performance metrics help detect potential system issues early.

Python and Pandas were used for data preprocessing and transformation.

Power BI dashboards enable clear visualization of server health.

This analysis supports better infrastructure management decisions.